**Figure 3a** | VO2max response in intervention versus control groups per exercise modality

| Author | Date | Study Design Type | Intervention Duration | Intervention | Indicator | INT | | | CON | | | Derivative Statistics | | | | 95% CI | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Mean | SD | Sample Size | Mean | SD | Sample Size | Pooled SD | SMD | SE | IV | Lower | Upper |
| Kohrt* | 1991 | C | 36-52 weeks | END | VO2max | 30.8 | 4.7 | 110 | 24.9 | 3.5 | 81 | 4.14 | 1.42 | 0.30 | 1.42 | 0.83 | 2.01 |
| Nybo | 2010 | C | 12 weeks | END | VO2max | 42.2 | 1.8 | 9 | 38.9 | 2.4 | 11 | 2.12 | 1.56 | 0.47 | 1.56 | 0.64 | 2.48 |
| Osei-Tutu & Campagna* | 2004 | RCT | 8 weeks | END | VO2max | 31.4 | 3.8 | 15 | 32.5 | 5.4 | 10 | 4.66 | -0.24 | 0.93 | -0.24 | -2.06 | 1.58 |
| Trapp | 2008 | RCT | 15 weeks | END | VO2max | 36.9 | 1.5 | 15 | 30.7 | 1.6 | 15 | 1.55 | 4.00 | 0.28 | 4.00 | 3.45 | 4.55 |
| Warburton | 2004 | RCT | 12 weeks | END | VO2max | 49.6 | 5.8 | 6 | 38.8 | 7 | 8 | 6.43 | 1.68 | 1.72 | 1.68 | -1.69 | 5.05 |
| Berger | 2005 | C | 6 weeks | END | VO2max | 40.3 | 5.3 | 8 | 35.4 | 8.3 | 7 | 6.96 | 0.70 | 1.80 | 0.70 | -2.83 | 4.23 |
| O-Donovan (higher intensity) | 2005 | RCT | 24 weeks | END | VO2max | 38.9 | 8.08 | 22 | 28.4 | 5.4 | 20 | 6.87 | 1.53 | 1.06 | 1.53 | -0.55 | 3.61 |
| O-Donovan (lower intensity) | 2005 | RCT | 24 weeks | END | VO2max | 35.8 | 6.3 | 22 | 28.4 | 5.4 | 20 | 5.87 | 1.26 | 0.91 | 1.26 | -0.52 | 3.04 |
| Hautala | 2006 | RCT (crossover) | 2 weeks | END | VO2max | 37 | 7 | 73 | 36 | 7 | 18 | 7.00 | 0.14 | 0.73 | 0.14 | -1.29 | 1.57 |
| Nybo | 2010 | C | 12 weeks | HIIT | VO2max | 41.4 | 2.2 | 8 | 38.9 | 2.4 | 11 | 2.30 | 1.09 | 0.53 | 1.09 | 0.05 | 2.13 |
| Trapp | 2008 | RCT | 15 weeks | HIIT | VO2max | 36.4 | 1.5 | 15 | 30.7 | 1.6 | 15 | 1.55 | 3.68 | 0.28 | 3.68 | 3.13 | 4.23 |
| Metcalfe* | 2012 | RCT | 6 weeks | HIIT | VO2max | 34.3 | 1.9 | 15 | 34.3 | 2.2 | 14 | 2.05 | -0.03 | 0.38 | -0.03 | -0.77 | 0.71 |
| Ziemann | 2011 | RCT | 6 weeks | HIIT | VO2max | 55.6 | 5.5 | 10 | 48.5 | 3.9 | 11 | 4.77 | 1.49 | 1.04 | 1.49 | -0.55 | 3.53 |
| Warburton | 2004 | RCT | 12 weeks | HIIT | VO2max | 47.3 | 12.7 | 6 | 38.8 | 7 | 8 | 10.25 | 0.83 | 2.74 | 0.83 | -4.54 | 6.20 |
| Berger | 2005 | C | 6 weeks | HIIT | VO2max | 42.5 | 8.9 | 8 | 35.4 | 8.3 | 7 | 8.61 | 0.83 | 2.22 | 0.83 | -3.52 | 5.18 |
| Astorino ** | 2018 | C | 3 weeks | HIIT | VO2max | 40.4 | 4.6 | 14 | 40.5 | 6.4 | 14 | 5.57 | -0.02 | 1.05 | -0.02 | -2.08 | 2.04 |
| Weatherwax (Standardised) | 2018 | RCT | 12 weeks | HIIT | VO2max | 26 | 4.2 | 20 | 27.7 | 4.6 | 8 | 4.40 | -0.39 | 0.83 | -0.39 | -2.02 | 1.24 |
| Weatherwax (Individualised) | 2018 | RCT | 12 weeks | HIIT | VO2max | 32.8 | 8.6 | 19 | 27.7 | 4.6 | 8 | 6.90 | 0.74 | 1.33 | 0.74 | -1.87 | 3.35 |
| Nybo | 2010 | C | 12 weeks | RET | VO2max | 37.9 | 3.3 | 8 | 38.9 | 2.4 | 11 | 2.89 | -0.35 | 0.66 | -0.35 | -1.64 | 0.94 |
| Hautala | 2006 | RCT (crossover) | 2 weeks | RET | VO2max | 36 | 7 | 73 | 36 | 7 | 18 | 7.00 | 0.00 | 0.73 | 0.00 | -1.43 | 1.43 |

**Legend**

**Notes**

** Pooled mean and SD calculated between male and female groups. Reported to 1 d.p. ** Derived from RAMP (incremental exercise test) data.*

**Abbreviations**

C = Case-control, RCT = Randomised controlled trial, END = Endurance, HIIT = High-intensity interval training, RET = Resistance training, INT = Intervention, SD = Standard deviation, CON = Control, SMD = Standardised mean difference, SE = Standard error, IV = Inverse variance, CI = Confidence interval

**Figure 3b** | VO2max response comparison between END versus HIIT and END versus RET

| Author | Date | Study Design Type | Intervention Duration | Intervention | Indicator | END | | | HIIT | | | Derivative Statistics | | | | 95% CI | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Mean | SD | Sample Size | Mean | SD | Sample Size | Pooled SD | SMD | SE | IV | Lower | Upper |
| Berger | 2005 | C | 6 weeks | END vs. HIIT | VO2max | 40.3 | 5.3 | 8 | 42.5 | 8.9 | 8 | 7.32 | -0.30 | 1.83 | -0.30 | -3.89 | 3.29 |
| Bonafiglia | 2016 | RCT | 3 weeks x 2 | END vs. HIIT | VO2max | 43.9 | 6.4 | 21 | 42.9 | 5.5 | 21 | 5.97 | 0.17 | 0.92 | 0.17 | -1.63 | 1.97 |
| Burgomaster * | 2008 | C | 6 weeks | END vs. HIIT | VO2max | 45 | 2 | 10 | 44 | 2 | 10 | 2.00 | 0.50 | 0.45 | 0.50 | -0.38 | 1.38 |
| Dunham | 2012 | RCT | 4 weeks | END vs. HIIT | VO2max | 34.3 | 6.7 | 7 | 36.5 | 6.7 | 8 | 6.70 | -0.33 | 1.73 | -0.33 | -3.72 | 3.06 |
| MacPherson | 2010 | C | 6 weeks | END vs. HIIT | VO2max | 49.5 | 8.3 | 10 | 52.2 | 6.5 | 10 | 7.45 | -0.36 | 1.67 | -0.36 | -3.63 | 2.91 |
| Matsuo | 2013 | RCT | 8 weeks | END vs. HIIT | VO2max | 41.9 | 5.6 | 14 | 42 | 6.8 | 14 | 6.23 | -0.02 | 1.18 | -0.02 | -2.33 | 2.29 |
| McKay | 2009 | RCT | 3 weeks | END vs. HIIT | VO2max | 46 | 6 | 6 | 48 | 4 | 6 | 5.10 | -0.39 | 1.47 | -0.39 | -3.27 | 2.49 |
| Nybo | 2010 | C | 12 weeks | END vs. HIIT | VO2max | 42.2 | 1.8 | 9 | 41.4 | 2.2 | 8 | 2.01 | 0.40 | 0.49 | 0.40 | -0.56 | 1.36 |
| Sandvel | 2012 | RCT | 8 weeks | END vs. HIIT | VO2max | 49.7 | 1.8 | 12 | 53.5 | 1.7 | 11 | 1.75 | -2.17 | 0.37 | -2.17 | -2.9 | -1.44 |
| Shepherd | 2013 | RCT | 6 weeks | END vs. HIIT | VO2max | 48.2 | 5 | 8 | 45.1 | 2.3 | 8 | 3.89 | 0.80 | 0.97 | 0.80 | -1.1 | 2.7 |
| Trapp | 2008 | RCT | 15 weeks | END vs. HIIT | VO2max | 36.9 | 1.5 | 15 | 36.4 | 1.5 | 15 | 1.50 | 0.33 | 0.27 | 0.33 | -0.2 | 0.86 |
| Warburton ** | 2004 | RCT | 12 weeks | END vs. HIIT | VO2max | 49.6 | 5.8 | 6 | 47.3 | 13 | 6 | 9.87 | 0.23 | 2.85 | 0.23 | -5.36 | 5.82 |
| Nybo | 2010 | C | 12 weeks | END vs. RET | VO2max | 42.2 | 1.8 | 9 | 37.9 | 3.3 | 8 | 2.66 | 1.62 | 0.64 | 1.62 | 0.37 | 2.87 |
| Hautala | 2006 | RCT (crossover) | 2 weeks | END vs. RET | VO2max | 37 | 7 | 73 | 36 | 7 | 73 | 7.00 | 0.14 | 0.58 | 0.14 | -1 | 1.28 |

**Legend**

**Notes**

** 6 week values obtained. ** Total (male and female) values incorporated.*

**Abbreviations**

C = Case-control, RCT = Randomised controlled trial, END = Endurance, HIIT = High-intensity interval training, RET = Resistance training, SD = Standard deviation, SMD = Standardised mean difference, SE = Standard error, IV = Inverse variance, CI = Confidence Interval

**Figure 3c** | Per-exercise modality comparison in response with alternative response indicators

| Author | Date | Study Design Type | Intervention Duration | Intervention | Indicator | END | | | HIIT | | | Derivative Statistics | | | | 95% CI | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Mean | SD | Sample Size | Mean | SD | Sample Size | Pooled SD | SMD | SE | IV | Lower | Upper |
| Kohrt * | 1991 | C | 36-52 weeks | END | HRrest | 76.1 | 12.0 | 110 | 78.4 | ### | 81 | 12.28 | -0.18 | 0.89 | -0.18 | -1.92 | 1.56 |
| Nybo | 2010 | C | 12 weeks | END | HRrest | 53 | 2 | 9 | 61 | 3 | 11 | 2.55 | -3.14 | 0.57 | -3.14 | -4.26 | -2.02 |
| Kohrt * | 1991 | C | 36-52 weeks | END | HRmax | 163.4 | 12.4 | 110 | 163.9 | ### | 81 | 11.73 | -0.05 | 0.85 | -0.05 | -1.72 | 1.62 |
| Nybo ** | 2010 | C | 12 weeks | END | HRmax | | | | | | | | | | | | |
| Lo † | 2011 | RCT | 24 weeks | END | HRmax | 195.1 | 7.5 | 10 | 192.1 | 10 | 10 | 8.90 | 0.34 | 1.99 | 0.34 | -3.56 | 4.24 |
| Warburton | 2004 | RCT | 12 weeks | END | HRmax | 188 | 12 | 6 | 185 | 6 | 8 | 9.49 | 0.32 | 2.54 | 0.32 | -4.66 | 5.3 |
| Berger | 2005 | C | 6 weeks | END | HRmax | 170 | 11 | 8 | 177 | 12 | 7 | 11.51 | -0.61 | 2.97 | -0.61 | -6.43 | 5.21 |
| Hautala | 2006 | RCT (crossover) | Two weeks | END | HRmax | 181 | 9 | 73 | 180 | 11 | 18 | 10.05 | 0.10 | 1.05 | 0.10 | -1.96 | 2.16 |
| Berger | 2005 | C | 6 weeks | END | LT | 5.2 | 1.1 | 8 | 4.6 | 0.6 | 7 | 0.89 | 0.68 | 0.23 | 0.68 | 0.23 | 1.13 |
| Warburton | 2004 | RCT | 12 weeks | END | PO | 291 | 71 | 6 | 229 | 39 | 8 | 57.28 | 1.08 | #### | 1.08 | -28.93 | 31.09 |
| Berger | 2005 | C | 6 weeks | END | PO | 194 | 41 | 8 | 174 | 50 | 7 | 45.72 | 0.44 | #### | 0.44 | -22.71 | 23.59 |
| Nybo | 2010 | C | 12 weeks | HIIT | HRrest | 52 | 2 | 8 | 61 | 3 | 11 | 2.55 | -3.53 | 0.58 | -3.53 | -4.67 | -2.39 |
| Weatherwax (Standardised) | 2018 | RCT | 12 weeks | HIIT | HRrest | 68.2 | 8 | 20 | 69.5 | 7.5 | 8 | 7.75 | -0.17 | 1.47 | -0.17 | -3.05 | 2.71 |
| Weatherwax (Individualised) | 2018 | RCT | 12 weeks | HIIT | HRrest | 68.1 | 11.4 | 19 | 69.5 | 7.5 | 8 | 9.65 | -0.15 | 1.86 | -0.15 | -3.8 | 3.5 |
| Nybo ** | 2010 | C | 12 weeks | HIIT | HRmax | | | | | | | | | | | | |
| Berger | 2005 | C | 6 weeks | HIIT | HRmax | 178 | 14 | 8 | 177 | 12 | 7 | 13.04 | 0.08 | 3.37 | 0.08 | -6.53 | 6.69 |
| Astorino ‡ | 2018 | C | 3 weeks | HIIT | HRmax | 182.5 | 11.8 | 14 | 182.7 | 7.2 | 14 | 9.77 | -0.02 | 1.85 | -0.02 | -3.65 | 3.61 |

**Legend**

**Notes**

** Pooled mean and SD calculated between male and female groups from "Final" datasets. Reported to 1 d.p. † Reported to 1 d.p.*

*** Omitted - measured submaximal HR at exertion with 9.5kmph training velocity. ‡ Derived from RAMP (incremental exercise test) data.*

**Abbreviations**

C = Case-control, RCT = Randomised controlled trial, END = Endurance, HIIT = High-intensity interval training, RET = Resistance training, HRrest = Heart rate at rest, Hrmax = Maximal heart rate, LT = Lactate threshould , PO = Power output , SD = Standard deviation, SMD = Standardised mean difference, SE = Standard error, IV = Inverse variance, CI = Confidence interval

*continued*

**Figure 3c** | Per-exercise modality comparison in response with alternative response indicators (cont.)

| Author | Date | Study Design Type | Intervention Duration | Intervention | Indicator | END | | | HIIT | | | Derivative Statistics | | | | 95% CI | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Mean | SD | Sample Size | Mean | SD | Sample Size | Pooled SD | SMD | SE | IV | Lower | Upper |
| Weatherwax (Standardised) | 2018 | RCT | 12 weeks | HIIT | HRmax | 164.9 | 15.1 | 20 | 170.1 | 11 | 8 | 13.21 | -0.39 | 2.50 | -0.39 | -5.29 | 4.51 |
| Weatherwax (Individualised) | 2018 | RCT | 12 weeks | HIIT | HRmax | 169.2 | 14.4 | 19 | 170.1 | 11 | 8 | 12.81 | -0.07 | 2.47 | -0.07 | -4.91 | 4.77 |
| Berger | 2005 | C | 6 weeks | HIIT | LT | 5.1 | 0.8 | 8 | 4.6 | 0.6 | 7 | 0.71 | 0.71 | 0.18 | 0.71 | 0.36 | 1.06 |
| Ziemann | 2011 | RCT | 6 weeks | HIIT | PO | 307 | 29 | 10 | 250 | 31 | 11 | 30.02 | 1.90 | 6.55 | 1.90 | -10.94 | 14.74 |
| Warburton | 2004 | RCT | 12 weeks | HIIT | PO | 305 | 67 | 6 | 229 | 39 | 8 | 54.82 | 1.39 | #### | 1.39 | -27.32 | 30.1 |
| Berger | 2005 | C | 6 weeks | HIIT | PO | 224 | 73 | 8 | 174 | **50** | **7** | 62.57 | 0.80 | #### | 0.80 | -30.85 | 32.45 |
| Astorino ‡ | 2018 | C | 3 weeks | HIIT | PO | 267 | 57 | 14 | 287 | 60 | 14 | 58.52 | -0.34 | #### | -0.34 | -22.02 | 21.34 |
| Nybo | 2010 | C | 12 weeks | RET | HRrest | 56 | 2 | 8 | 61 | 3 | 11 | 2.55 | -1.96 | 0.58 | -1.96 | -3.1 | -0.82 |
| Nybo ** | 2010 | C | 12 weeks | RET | HRmax | | | | | | | | | | | | |
| Lo † | 2011 | RCT | 24 weeks | RET | HRmax | 196.2 | 9.2 | 10 | 192.1 | 10 | 10 | 9.66 | 0.42 | 2.16 | 0.42 | -3.81 | 4.65 |
| Hautala | 2006 | C | 2 weeks | RET | HRmax | 180 | 10 | 73 | 180 | 11 | 18 | 10.51 | 0.00 | 1.10 | 0.00 | -2.16 | 2.16 |

**Legend**

**Notes**

** Pooled mean and SD calculated between male and female groups from "Final" datasets. Reported to 1 d.p. † Reported to 1 d.p.*

*** Omitted - measured submaximal HR at exertion with 9.5kmph training velocity. ‡ Derived from RAMP (incremental exercise test) data.*

**Abbreviations**

C = Case-control, RCT = Randomised controlled trial, END = Endurance, HIIT = High-intensity interval training, RET = Resistance training,
HRrest = Heart rate at rest, Hrmax = Maximal heart rate, LT = Lactate threshould , PO = Power output , SD = Standard deviation,
SMD = Standardised mean difference, SE = Standard error, IV = Inverse variance, CI = Confidence interval